



Batterjee Medical College
For Science and Technology

Diffuse Parenchymal Lung Diseases

An approach to a Case of Dyspnea

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Main Subtopics



- Overview
- DPLDs Classification
- DLPDs Manifestations
- IPF
- COP
- Radiation Pneumonitis
- Sarcoidosis
- Pneumoconiosis
- Misc. DPLDs

Learning Objectives

Knowledge:

- Define restrictive lung diseases and classify DPLDs into idiopathic, exposure-related and systemic disease-associated categories.
- Describe the mechanisms of alveolar tissue injury, fibroblast proliferation, and matrix deposition leading to pulmonary fibrosis, as well as specific processes like particle phagocytosis in pneumoconioses and noncaseating granuloma formation in sarcoidosis.
- Identify classic signs and symptoms and extrapulmonary manifestations along with potential complications like pulmonary hypertension and increased risk of malignancy or infections in DPLS and sarcoidosis.
- Detail characteristic diagnostic findings across pulmonary function tests, imaging, and targeted pharmacologic or non-pharmacologic therapies.

Skills:

- Analyze clinical manifestations, lab results, pulmonary function tests, and radiological imaging to differentiate between various intrinsic and extrinsic restrictive lung disorders.
- Formulate evidence-based diagnostic and therapeutic management plans tailored to specific DPLDs, including sarcoidosis, identifying drug-induced causes and selecting appropriate options.

Value:

Demonstrate a professional attitude and adherence to training site rules, regulations, and clinical protocols while caring for patients presenting with complex respiratory disorders.

Overview

RESTRICTIVE LUNG DISEASES

↳ **INHALATION FILLS LUNGS** **LESS** than **NORMAL** HIGH YIELD

DIFFUSE PARENCHYMAL LUNG DISEASES

(INTERSTITIAL LUNG DISEASES)

↳ LUNG TISSUE DAMAGED

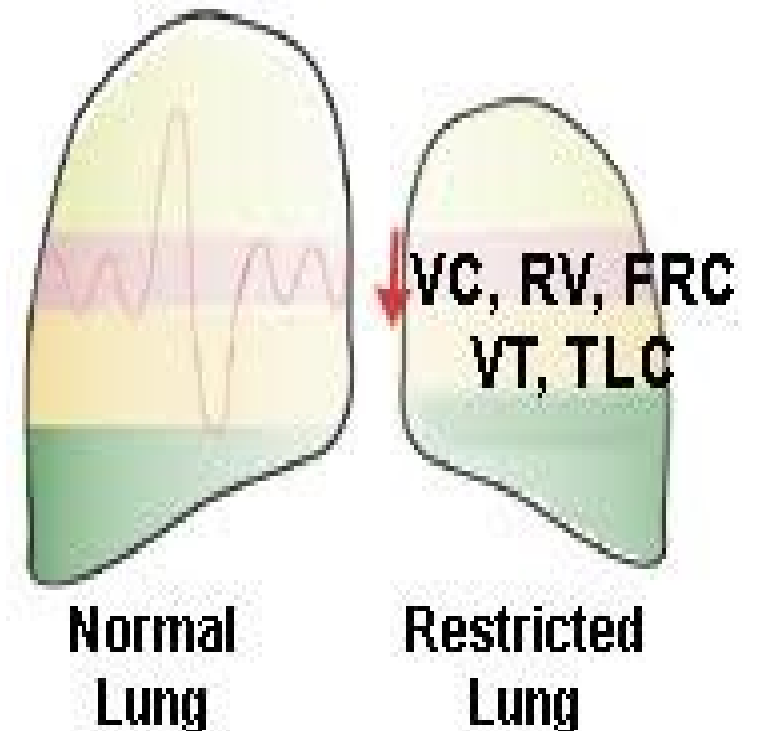
↳ **FIBROTIC, RIGID LUNG w/ ↓ COMPLIANCE & ↑ RECOIL**



↓ LUNG VOLUMES HIGH YIELD

EXTRAPULMONARY LUNG DISEASES

Restrictive Lung Disorders



Overview

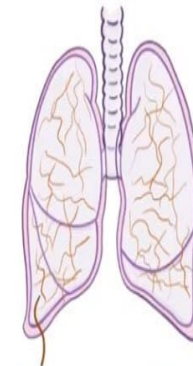
Intrinsic causes (parenchymal diseases)

- Interstitial lung disease (diffuse parenchymatous lung disease)
- Sarcoidosis
- Pneumoconioses
- ARDS
- Idiopathic pulmonary fibrosis
- Hypersensitivity pneumonitis (extrinsic allergic alveiolitis)
- Granulomatosis with polyangiitis
- Radiation-induced lung injury
- Drug-induced interstitial lung disease
- Alveolar (e.g., pneumonia, pulmonary edema, hemorrhage)

Extrinsic causes (i.e., extrapulmonary conditions that change the mechanics of respiration)

- Diseases of the pleura (chronic pleural effusion, pneumothorax)
- Deformities of the thorax/mechanical limitation (kyphoscoliosis, ankylosing spondylitis, pregnancy, ascites)
- Respiratory muscle weakness (phrenic nerve palsy, myasthenia gravis)

INTERSTITIAL RESTRICTIVE LUNG DISEASE



FIBROTIC & FIRM

↓ TOTAL LUNG CAPACITY

↓ FVC

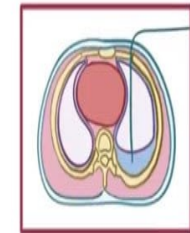
↓ FEV1

NORMAL or ↑ FEV1:FVC

EXTRA-PULMONARY RESTRICTIVE LUNG DISEASE



ABNORMALITIES
IN CHEST WALL
OR
DIAPHRAGM



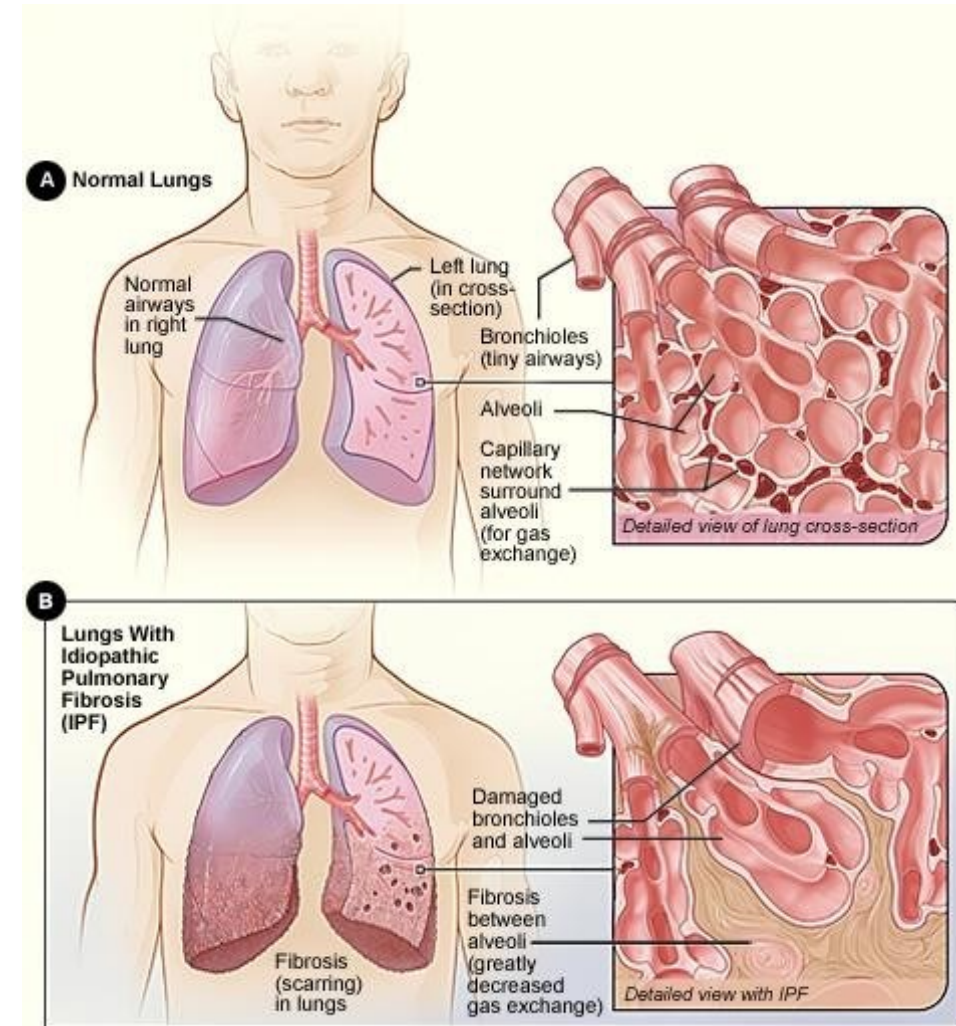
PLEURAL
EFFUSIONS

AND NEUROMUSCULAR DISEASES

Diffuse Parenchymal Lung Diseases

Definition:

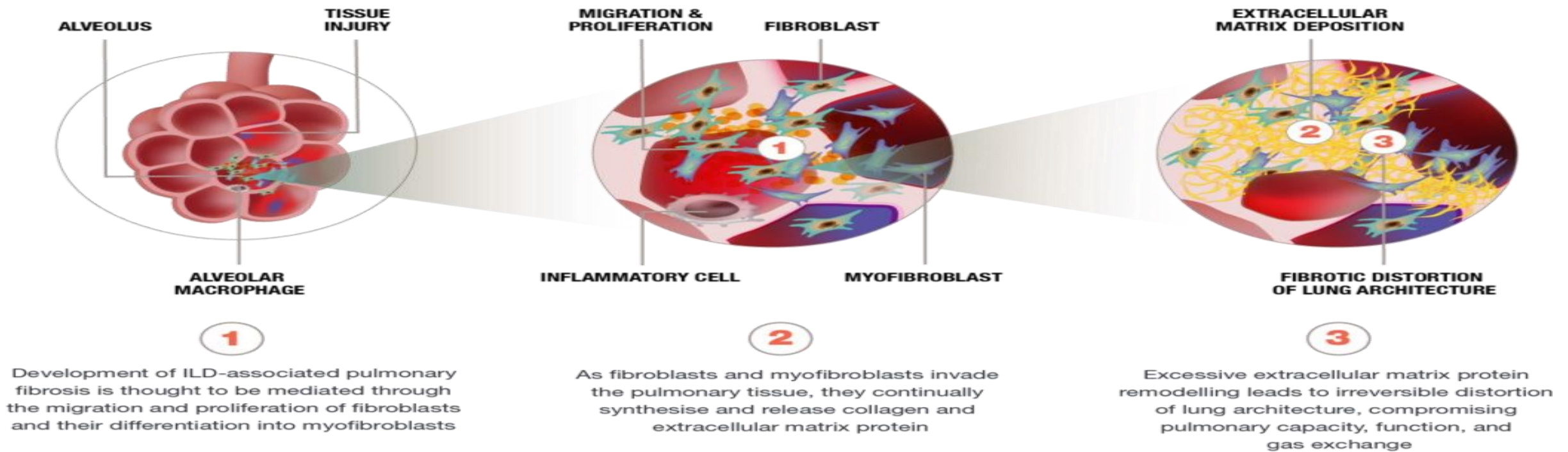
- A diverse group of rare, highly morbid pulmonary disorders
- Characterized by progressive inflammatory process involving the lungs
- Results in widespread scarring (fibrosis)
- Previously known as interstitial lung diseases (ILD)



Diffuse Parenchymal Lung Diseases

Pathophysiology:

- **General:** All types of ILDs share the same basic pathophysiology.
- Repeated cycles of tissue injury in the lung parenchyma with aberrant wound healing → collagenous fibrosis → remodeling of the pulmonary interstitium



Diffuse Parenchymal Lung Diseases

Clinical Features:

- Progressive dyspnea (first on exertion than on rest)
- Persistent nonproductive cough
- Bibasilar, inspiratory crackles (velcro-like(fine crackles))
- Fatigue
- Clubbing
- Fever (hypersensitivity pneumonitis & sarcoidosis)
- Findings suggestive of connective tissue disease, sarcoidosis, or vasculitis.

Advanced disease

- Digital clubbing due to chronic hypoxia
- Cyanosis
- Loud inspiratory wheeze

Interstitial Lung Diseases

Diffuse Parenchymal Lung Diseases

INFLAMMATION AND/OR FIBROSIS IN LUNG TISSUE.

RESTRICTIVE —

✓ REDUCED TLC, FVC, FEV1

SIGNS & SYMPTOMS

✓ PROGRESSIVE DYSPNEA

ON EXERTION -> AT REST

✓ DRY COUGH

✓ DIGIT CLUBBING

✓ LUNG OPACITIES ON CXR

COMPLICATIONS

✓ PULM. ARTERIAL HYPERTENSION

DIAGNOSIS

✓ DIFFICULT DUE TO NON-SPECIFIC,
OVERLAPPING SIGNS AND SYMPTOMS.

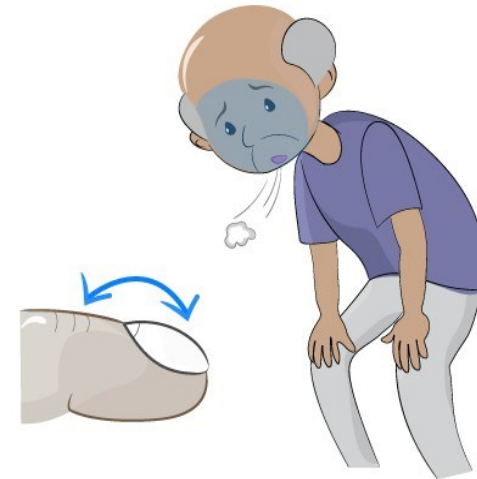
✓ IMAGING, PFTs, CLINICAL DATA, HISTOPATHOLOGY.

TREATMENT

✓ TARGET UNDERLYING DISEASE

✓ ANTI-FIBROTIC DRUGS

EX: PIRFENIDONE



Diffuse Parenchymal Lung Diseases

Diagnostics:

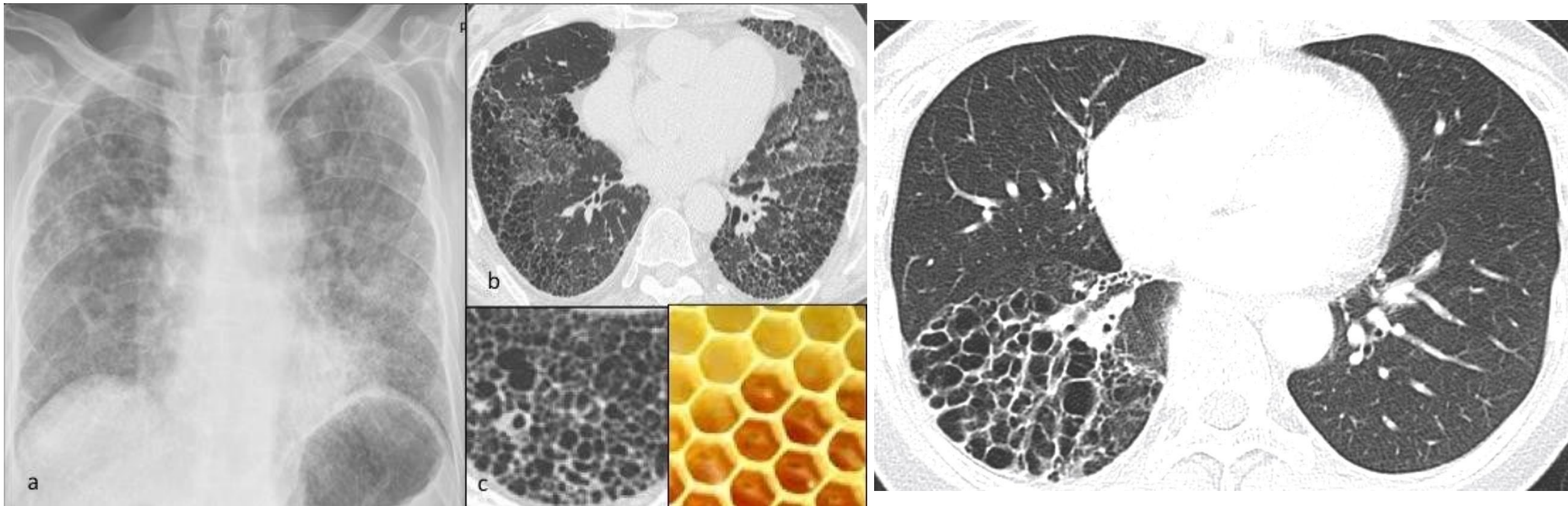
CXR

Diffuse opacification
(reticular, reticulonodular, ground glass, honeycombing)



Diffuse Parenchymal Lung Diseases

Diagnostics: Chest CT

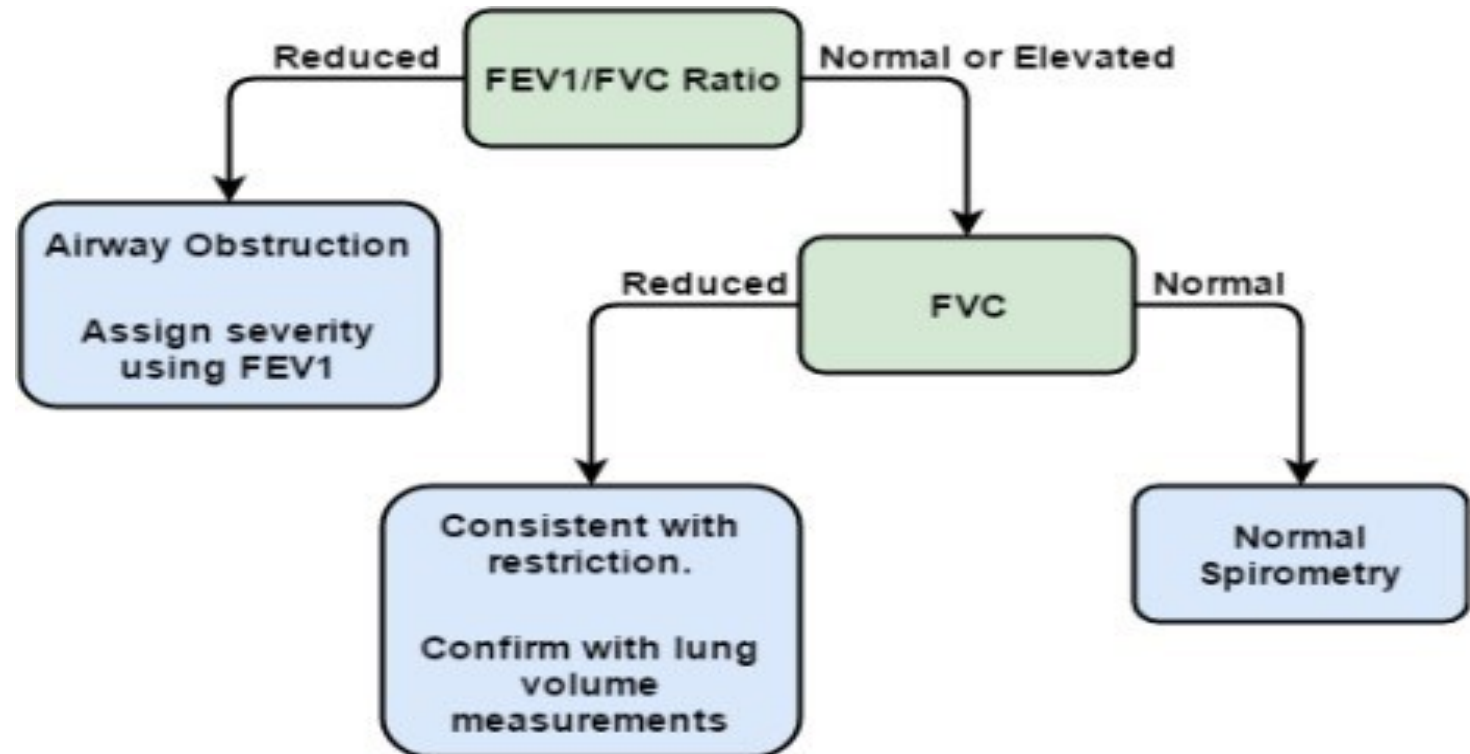


Diffuse Parenchymal Lung Diseases

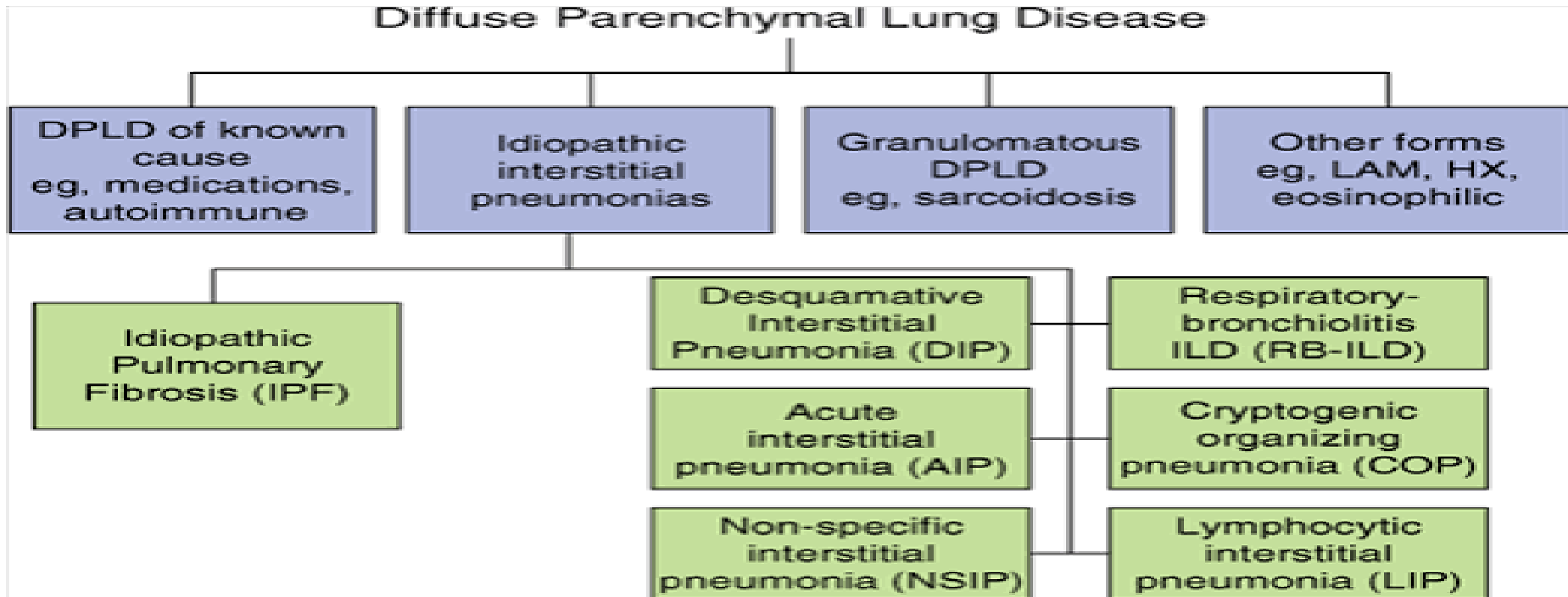
Diagnostics:

PFTs

- A restrictive pattern (FEV1/FVC ratio is normal or $\uparrow$)
- $\downarrow$ All lung volumes
- $\downarrow$ diffusing capacity (DLCO)



Diffuse Parenchymal Lung Diseases



Source: McKean S, Ross JJ, Dressler DD, Brotman DJ, Ginsberg JS: *Principles and Practice of Hospital Medicine*: www.accessmedicine.com

Diffuse Parenchymal Lung Diseases

Idiopathic pulmonary fibrosis (IPF) : most common type of ILD, characterized by irreversible pulmonary fibrosis and impaired pulmonary function.

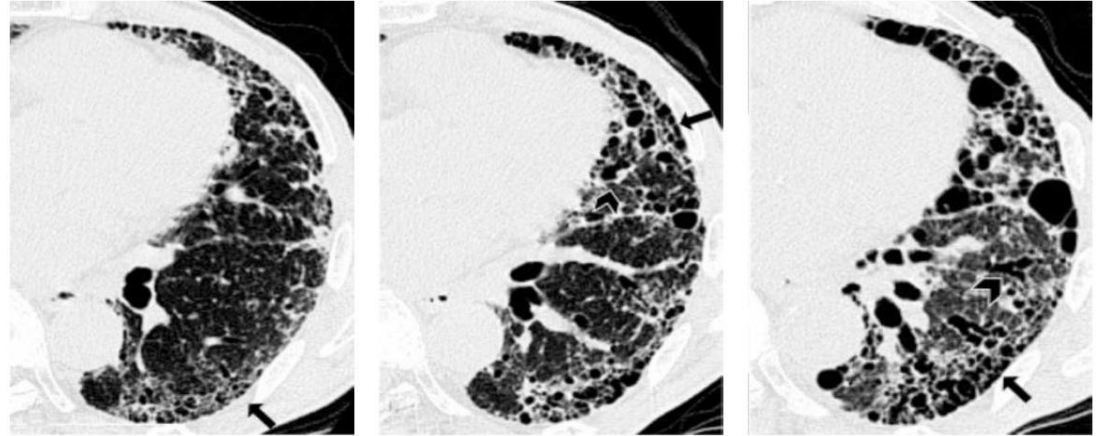
	SYMPTOMS/EXAM	DIAGNOSIS	TREATMENT
Idiopathic pulmonary fibrosis (IPF)—the most common ILD	Affects middle-age or older patients (>50 years); presents with chronic, nonproductive cough and gradually worsening dyspnea over many months Not acutely ill and do not present with fever; very slow progression Exam reveals bibasilar inspiratory crackles (Velcro-like), restrictive lung disease ($\downarrow$ FVC, TLC, FRC), and $\downarrow$ gas exchange (D_{LCO})	CXR with diffuse infiltrates (nonspecific) Lung biopsy reveals usual interstitial pneumonia; HRCT shows bibasilar, reticular patchiness with honeycombing and bronchiectasis in subpleural areas	Has a poor prognosis despite immunosuppressive therapy; lung transplantation can improve survival and quality of life
		 A B C	

FIGURE 4.9. Idiopathic pulmonary fibrosis (IPF). HRCT findings of a 78-year-old man with IPF who died 20 months after initial diagnosis. Lower left lobe shows fibrosing changes at baseline (A), 6 months (B), and 12 months (C). Changes include subpleural predominant interstitial fibrosis, traction bronchiectasis (arrowheads), and honeycombing (arrows). The overall HRCT fibrosis score at the baseline, 6 months, and 12 months were 151.3, 162.5, and 185.8, respectively. (Reproduced with permission from USMLE-Rx.com.)

Cryptogenic Organizing Pneumonia

Cryptogenic organizing pneumonia (COP): a rare, type of ILD characterized by inflammation of the bronchioles, alveolar ducts, and alveolar walls

Cryptogenic organizing pneumonia

Acutely ill over days to weeks, and mimics community-acquired pneumonia symptoms and may have initial **flu-like** symptoms (eg, fever, fatigue, nonproductive cough); dyspnea on exertion and weight loss are also common

50% are idiopathic, 50% are associated with toxic fume inhalation, certain drug exposures, infections, connective tissue disorders, immunodeficiencies, radiation

CXR and CT show bilateral **patchy** opacities with normal lung volumes
Lung biopsy reveals organizing pneumonia but is not diagnostic

Responsive to corticosteroids



Diffuse Parenchymal Lung Diseases

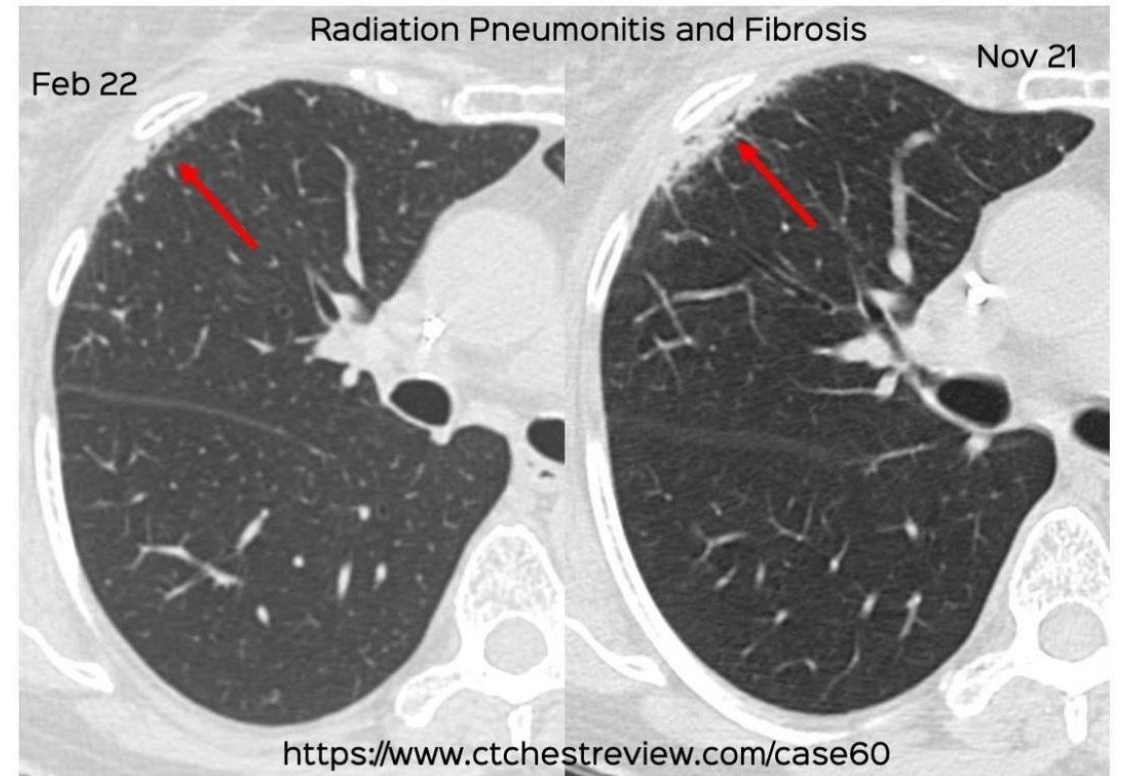
Acute interstitial pneumonia (AIP): (Hamman-Rich syndrome) a severe, acute ILD that can rapidly progress to respiratory failure

Nonspecific interstitial pneumonia (NSIP): a type of ILD associated with connective tissue diseases (e.g., systemic sclerosis), HIV infection, and hypersensitivity pneumonitis

Can be differentiated histologically

Radiation Pneumonitis

- 5% to 15% of patients who undergo thoracic irradiation
- Acute (1 to 6 months after irradiation)
- chronic (1 to 2 years after irradiation)
- Alveolar thickening and pulmonary fibrosis
- low-grade fever, cough, chest fullness, dyspnea, pleuritic chest pain, hemoptysis
- CT scan is the best study
- The treatment of choice is corticosteroids



Treatment

Treatment of IPF

•Pharmacological therapy

- Antifibrotic agents may reduce mortality and acute exacerbations and slow the decline in FVC.
 - Pirfenidone: inhibits TGF- β -stimulated collagen synthesis
 - Nintedanib: inhibits tyrosine kinases that target fibrogenic growth factors

•Lung transplantation: Refer for evaluation at the time of diagnosis.

Do not start patients with IPF on immunosuppressive therapy; immunomodulation may worsen outcomes.

Treatment of non-IPF ILD

•Pharmacological therapy: depending on the cause of ILD

- Treat the underlying condition
 - Antibiotics if bacterial interstitial pneumonia is suspected
- Consider corticosteroids and/or systemic immunomodulators.

•Lung transplantation

Sarcoidosis

General Characteristics:

- Chronic systemic granulomatous disease characterized by noncaseating Granulomas
- often involving multiple organ systems
- Etiology unknown (genetic , environmental exposure eg. Beryllium, infections)
- Occurs most often in African American women
- occur in individuals <40 years
- Sarcoidosis carries a good prognosis in majority of patients

Sarcoidosis

- a disease with highly variable manifestations, ranging from asymptomatic to progressive, relapsing disease.

- sudden onset with spontaneous remission in approx. two-thirds of patients
- Approx. one-third of patients develop progressive chronic disease.

- Most common presenting signs and symptoms

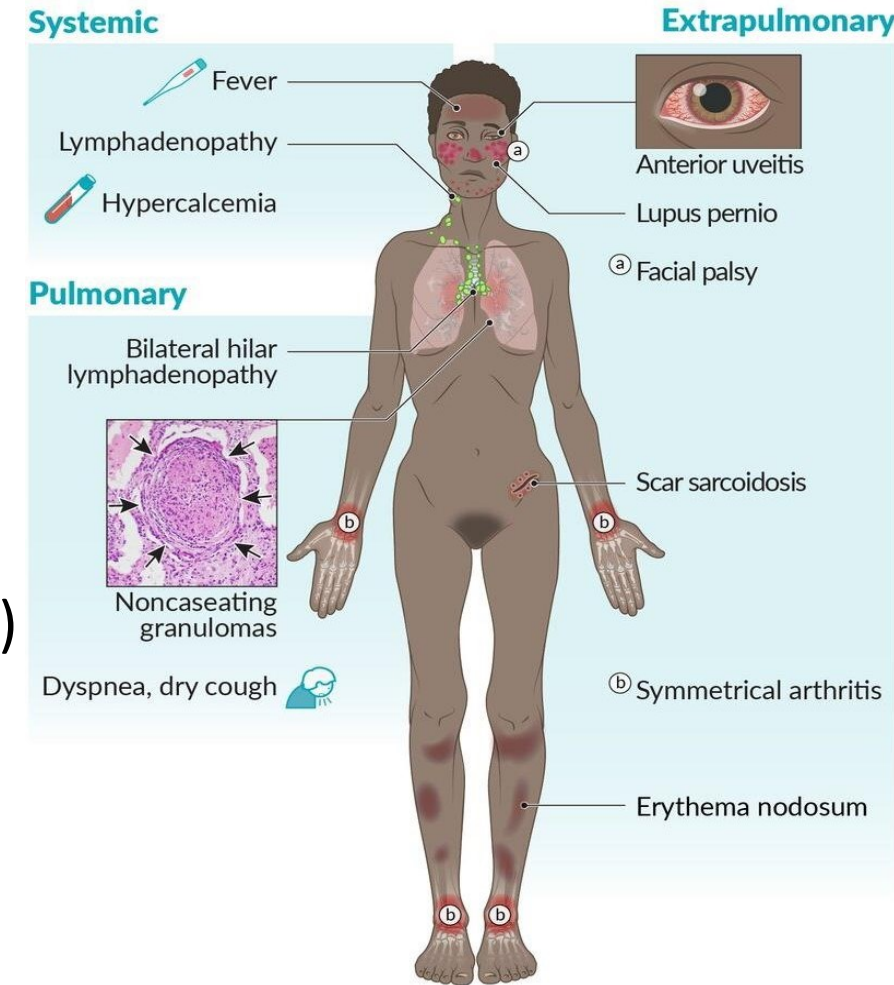
- Often asymptomatic (in early stages)
- General: fever, malaise, lack of appetite, weight loss, lymphadenopathy
- **Pulmonary:** dyspnea, cough, chest pain, rales on auscultation
- Hypercalcemia in sarcoidosis is primarily caused by the dysregulated production of 1,25-dihydroxyvitamin D (calcitriol) by macrophages within granulomas.
- Extrapulmonary: **arthritis, erythema nodosum, anterior uveitis**
- **Cutaneous sarcoidosis:** in approx. 30% of affected individuals
- Erythema nodosum: most common nonspecific cutaneous manifestation
- **Lupus pernio**, A pathognomonic manifestation of sarcoidosis characterized by extensive, violaceous skin plaques on the nose, cheeks, chin, and/or ears
- Butterfly rash as seen in systemic lupus erythematosus



Sarcoidosis

Clinical Features:

- Constitutional symptoms
- Erythema nodosum
- Polyarthralgias
- Hilar lymphadenopathy
- Liver disease, splenomegaly
- Cardiomyopathies
- Neurosarcoidosis
- Cranial nerve palsy (unilateral or bilateral facial nerve palsy)
- Diabetes insipidus
- Hypopituitarism
- Meningitis
- Peripheral neuropathy & Myopathy
- Kidneys: acute interstitial nephritis, nephrocalcinosis, nephrolithiasis



Sarcoidosis

Diagnostics: Sarcoidosis is a diagnosis of exclusion

- Biopsy (noncaseating granulomas)
- CXR
- HRCT
- LFTs, ↑ calcium, BUN/creatinine
- Urine analysis (hypercalcuria)
- Angiotensin-converting enzyme
- Ophthalmologic exam
- ECG
- Spirometry

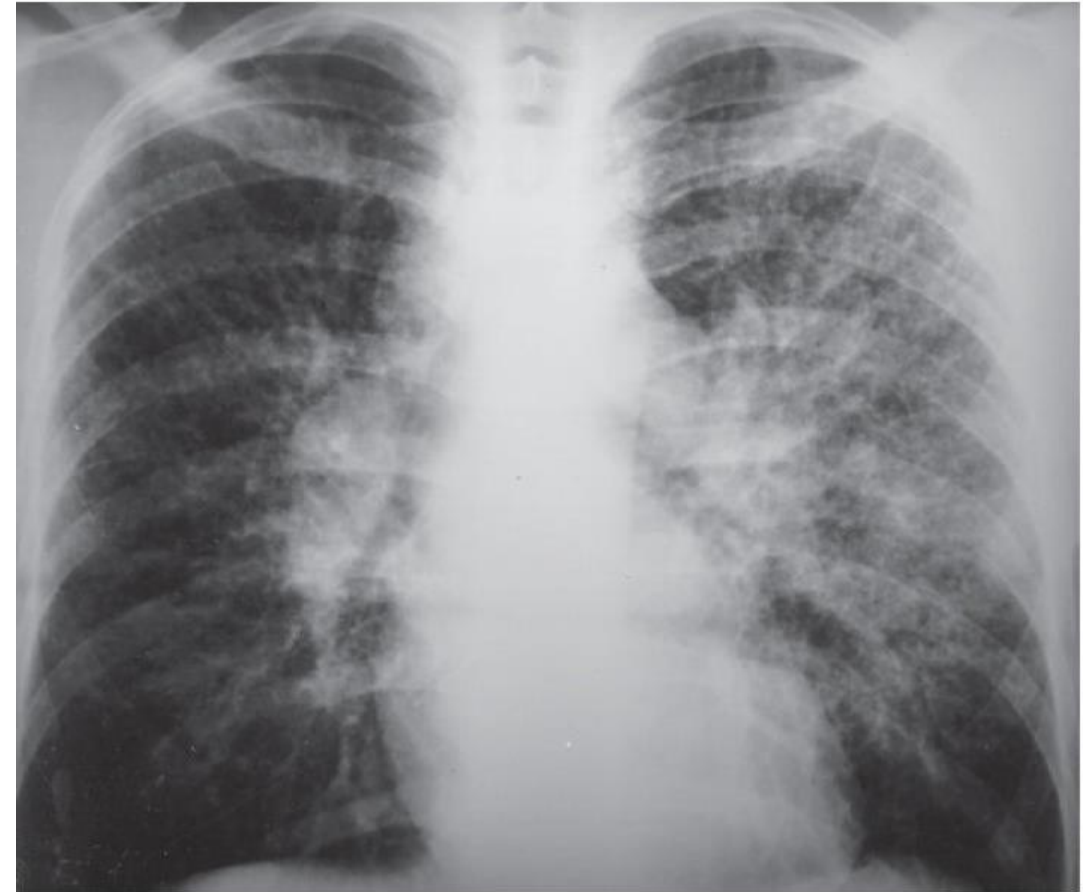
TABLE 4.16. Stages of Sarcoidosis

	CXR FINDINGS	SPONTANEOUS RESOLUTION WITHOUT TREATMENT
Stage 0	Normal	—
Stage 1	Hilar lymphadenopathy (see Figure 4.11)	60%-80%
Stage 2	Hilar lymphadenopathy and reticulonodular opacities	50%-60%
Stage 3	Reticulonodular opacities (no lymphadenopathy)	<30%
Stage 4	Fibrotic changes	—

Typically shows a restrictive pattern, Obstructive and mixed patterns may also be seen.

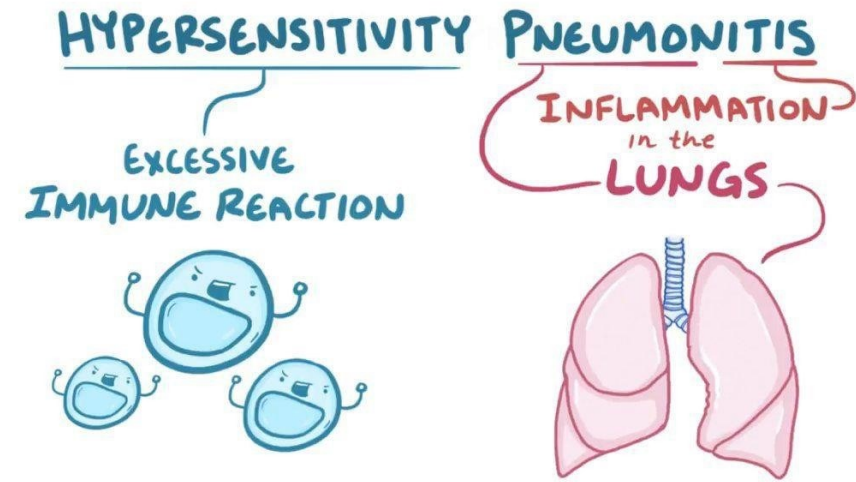
- Diffusing capacity of the lung for carbon monoxide (DLCO) is decreased.
- **Treatment:**
- Steroids. consider adding DMARD

Sarcoidosis



Hypersensitivity Pneumonitis

CATEGORY	PRESENTATION
OCCUPATIONAL/EXPOSURE	
Hypersensitivity pneumonitis (organic inhalations)	Caused by an allergic reaction to inhaled organic agents Acute symptoms: Fever, chills, cough, dyspnea 4-8 hours after exposure; inspiratory crackles; ground-glass opacities on CT Chronic symptoms: Gradual cough and dyspnea, malaise, weight loss; diffuse fibrosis on biopsy Treatment: Avoid common inciting agents such as mold, dust, bird droppings, aerosolized/humidified water (hot tubs), and pesticides; give corticosteroids if acute and severe



Quick HIT

Some causes of hypersensitivity pneumonitis:

- Farmer's lung (moldy hay)
- Bird-breeder's lung (avian droppings)
- Air-conditioner lung
- Bagassosis (moldy sugar cane)
- Mushroom worker's lung (compost)

Pneumoconiosis

• **Pneumoconioses:** a group of restrictive interstitial lung diseases caused by the inhalation of certain substances (mainly dusts) that often affect miners and agricultural workers

Asbestosis	• Airborne asbestos fibers	• Asbestos miners and millers, Brake linings and insulation manufactures, Ship construction workers, Firefighters, Plumbers
Silicosis	• Crystalline silica dust	• Miners, Workers of quarries, pottery production, sandblasting, glass manufacturers, Construction workers
Byssinosis	• Mainly dust from raw cotton • Less commonly: flax, jute, hemp, and sisal dust	• Textile industry workers

- **Pneumoconiosis:** inhalation of dust particles → phagocytosis by alveolar macrophages → destruction of alveolar macrophages, inflammatory reaction → scarring, granuloma formation

Pneumoconiosis

Asbestosis:

- Characterized by diffuse interstitial fibrosis of the lung caused by inhalation of asbestos fibers predilection for lower lobes.
- Develops insidiously many years (>15 to 20 years) after exposure.
- **Increased risk of bronchogenic carcinoma (smoking is synergistic) and malignant mesothelioma.**
- Diagnosis made based on clinical findings and history of exposure to asbestos.
- CXR shows hazy infiltrates with bilateral linear opacities and may show pleural plaques (especially in lower lung regions).
- No specific treatment is available.

Pneumoconiosis

Silicosis:

- **Localized** and **nodular** peribronchial fibrosis (upper lobes more common).
- Can be acute (massive exposure leading to rapid onset and death), or chronic (symptoms years after exposure—up to 15 years or longer).
- Associated with an increased risk of TB.
- **Sources include mining, stone cutting, and glass manufacturing.**
- Exertional dyspnea is the main symptom; cough with sputum is also seen.
- There are restrictive pulmonary function abnormalities.
- Treatment is supportive and involves removal from exposure to silica.

Quick

HIT



Classic CXR findings in environmental lung disease:

- Asbestosis: pleural plaques
- Silicosis: “eggshell” calcifications

Pneumoconiosis

DIAGNOSIS

* CHEST X-RAY

- ↳ NON-SPECIFIC INTERLACING WEB-LIKE RETICULAR INFILTRATES
 - ↳ SPHERICAL NODULAR INFILTRATES
- RETICULONODULAR PATTERN



* high resolution computed tomography → HRCT SCAN

- ↳ BILATERAL HILAR ADENOPATHY
 - ↳ SARCOIDOSIS
- ↳ UPPER LOBE CALCIFIED NODULES, called "EGGSHELL CALCIFICATIONS"
 - ↳ SILICOSIS
- ↳ LUNG PARENCHYMA → HONEYCOMB
 - ↳ IDIOPATHIC PULMONARY FIBROSIS
- ↳ PLEURAL EFFUSIONS
 - ↳ CONNECTIVE TISSUE DISEASES
- ↳ LINEAR PLEURAL PLAQUE
 - ↳ ASBESTOSIS

Pneumoconiosis

TREATMENT

- * DEPENDS on UNDERLYING CAUSE
- * OCCUPATIONAL LUNG DISEASE
 - ↳ AVOID OCCUPATIONAL ENVIRONMENT
- * QUIT SMOKING



ASBESTOSIS

- * FOLLOWED up for BRONCHOGENIC ADENOCARCINOMA or MESOTHELIOMA
 - ↳ especially if EXPOSED > 30 YRS
 - ↳ BRONCHOGENIC ADENOCARCINOMA is most COMMON TYPE of LUNG CANCER associated w/ ASBESTOS



Pneumoconiosis

TREATMENT



SILICOSIS

- * SUSCEPTIBLE to TUBERCULOSIS

↳ SILICA IMPAIRS FORMATION of PHAGOLYSOSOME
↳ MYCOBACTERIUM TUBERCULOSIS SURVIVES w/in CELL

- * SCREENED ANNUALLY w/ TUBERCULIN SKIN TEST

BERYLLIOSIS

- * CORTICOSTEROIDS
- * AVOIDING EXPOSURE to BERYLLIUM

HYPERSENSITIVITY PNEUMONITIS

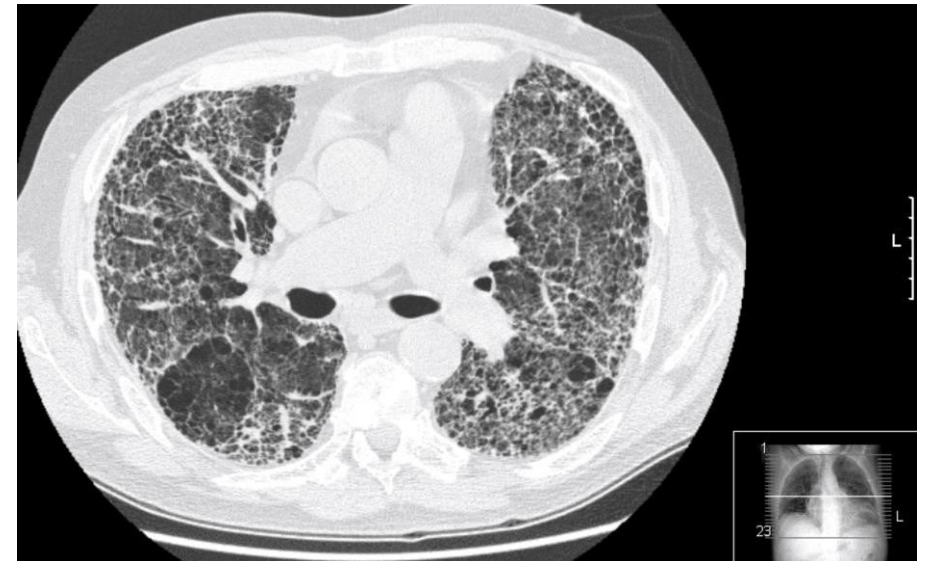
- * REMOVING the OFFENDING AGENT
- * SEVERE CASES → CORTICOSTEROIDS



Case Scenario 1

A 55-year-old woman comes to the physician because of an 8-month history of persistent dry cough and shortness of breath with mild exertion. She reports feeling too weak to leave the house on most days. She has rheumatoid arthritis and hypertension. She used to work as a hotel receptionist but quit her job a year ago because of worsening joint pain. She smoked a pack of cigarettes daily for 27 years but quit 2 years ago. Her medications include methotrexate, folic acid, and enalapril. Her temperature is 36.1°C (97°F), pulse is 98/min, respirations are 18/min, and blood pressure is 148/82 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 96%. Physical examination shows an enlargement of the fingertips with increased curving of the nails and bilaterally enlarged metacarpophalangeal joints. Inspiratory crackles are heard over both lung bases. A high-resolution CT scan of the chest is shown. The diffusing capacity of the lung (DL_{CO}) is decreased. Which of the following is the most likely diagnosis?

- a. Caplan syndrome
- b. Emphysema
- c. Pleural effusion
- d. Pulmonary edema
- e. Pulmonary hypertension
- f. Interstitial lung disease



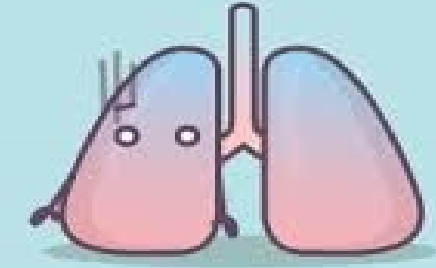
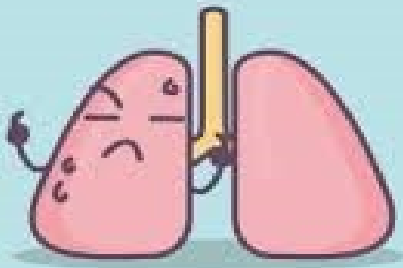
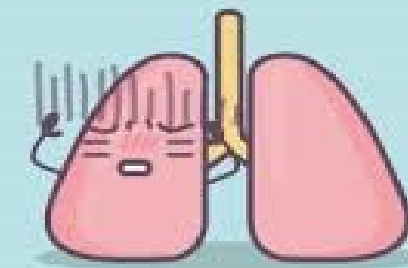
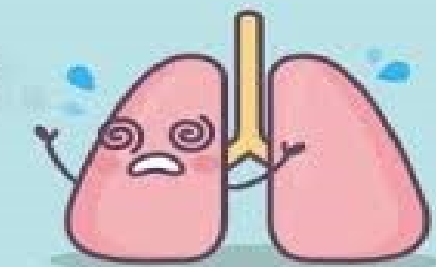
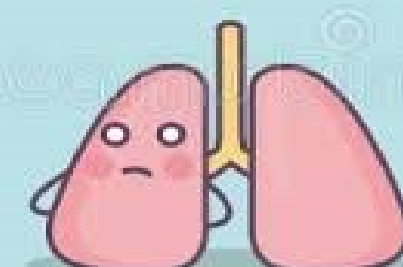
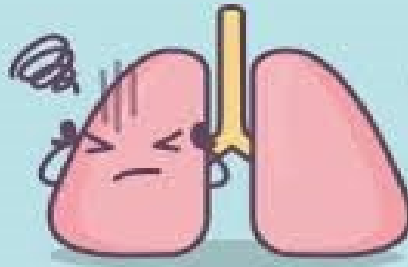
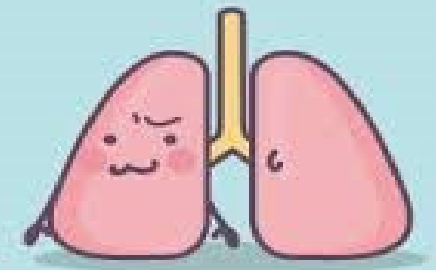
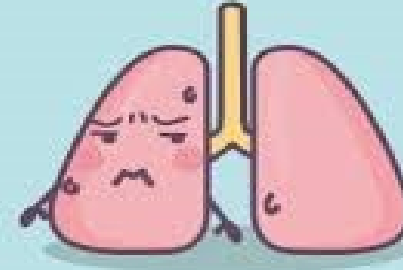
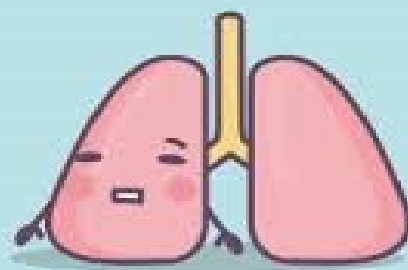
Case Scenario 2

A 35-year-old woman comes to the physician because of a dry cough and worsening shortness of breath with exertion for the past 6 months. She used to go running three times each week but had to stop because of decreased exercise tolerance and pain in the bilateral ankles. Two months ago, she was in Nigeria for several weeks to visit her family. She is allergic to cats and pollen. She has smoked one pack of cigarettes daily for the past 17 years. Her vital signs are within normal limits. Examination shows multiple 1.5- to 2-cm, nontender lymph nodes in the axillae. A few crackles are heard on auscultation of the chest. Her serum calcium concentration is 11.7 mg/dL. An x-ray of the chest shows hilar enlargement bilaterally and reticular opacities in both upper lungs. Which of the following is the most likely cause of these findings?

- a. Necrotizing inflammation
- b. Granulomatous inflammation
- c. Neoplastic transformation
- d. Viral infection
- e. Air trapping
- f. Bronchial hyperresponsiveness

Questions?

Confused Lung



References & further reading

- Step-up to medicine 5th edition
- Harrison's Principles of Internal Medicine 20th edition
- UpToDate
- Osmosis
- Amboss

Thank You